

HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps

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Video :



Motivation

Can a simple push or spoken command convey high-level intent in robot navigation tasks without menus, tablets, or teleoperation?

Challenges:

Short interactions are ambiguous
Many goals may lie in one push direction
Cues give a direction, not a full trajectory

Key Idea and mathematical formulation

We model human intent as a dynamical system and infer it from sparse cues.

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x} - \mathbf{x}^*)$$

- $\mathbf{x}^* \rightarrow$ where to go(goal)
- $\mathbf{A} \rightarrow$ how to get there(motion style)

Each particle is one candidate $(\mathbf{x}^*, \mathbf{A})$. Cues reweight the belief:

Push $\rightarrow$ velocity alignment with the particle's DS

Voice / gesture $\rightarrow$ bearing alignment with the commanded direction

Experiment setup

SBlimp: Crazyflie 2.1 with tilted rotors on a Mylar helium balloon

OptiTrack motion capture; commands via crazyswarm
Only the velocity controller runs onboard

Sensing the human:

Pushes: detected from mocap when speed exceeds the threshold velocity

Voice: laptop mic $\rightarrow$ Vosk (straight / right / left / back)

Gesture: webcam $\rightarrow$ MediaPipe Hands (pointing ray)

Particle filter in NumPy on CPU

Experiment Protocol :

5 participants $\times$ 60 trials = 300 trials

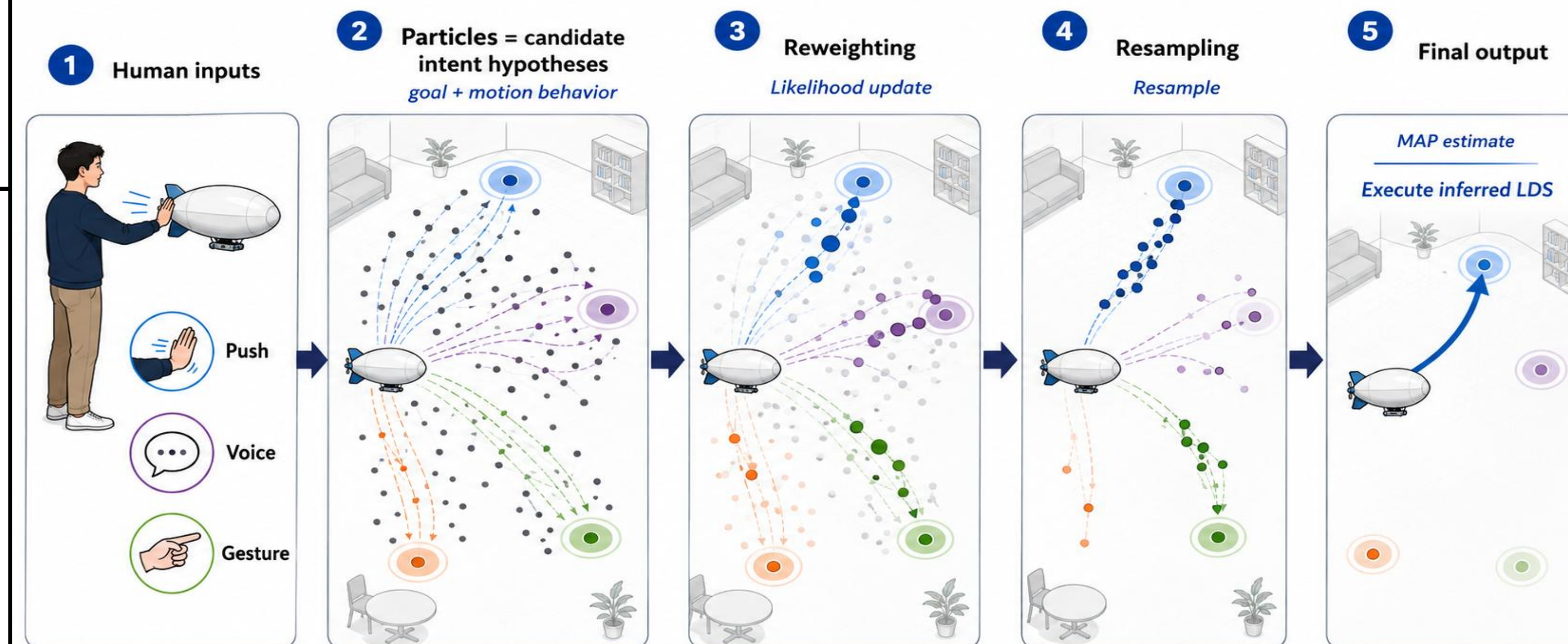
8 candidate goals, one chosen at random per trial, unknown to the robot

3 conditions: push only / voice only / combined

Success = reaches the correct goal in under 6 interactions

Uniform scene prior, so belief comes entirely from human cues

Particle filter for intent inference



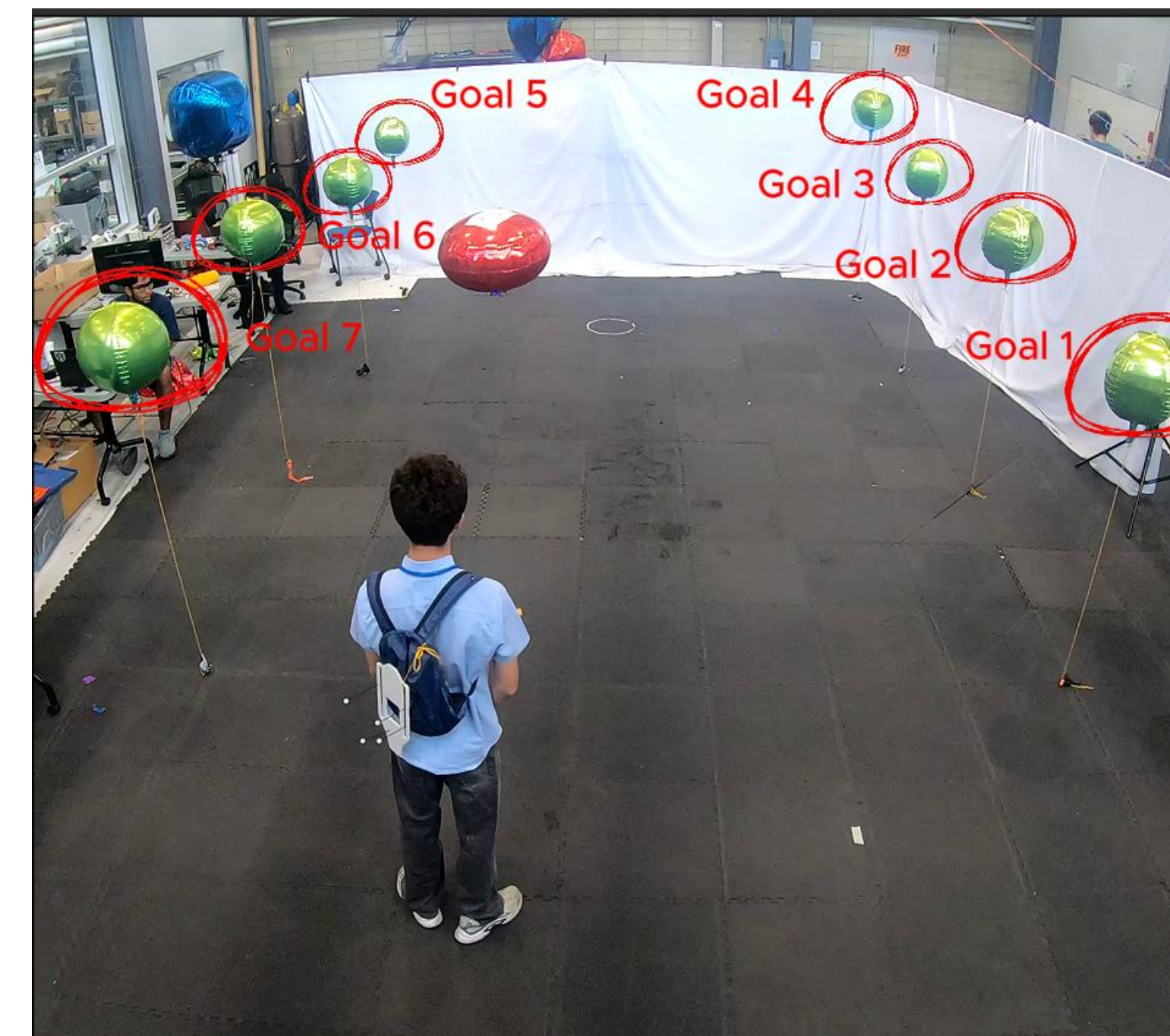
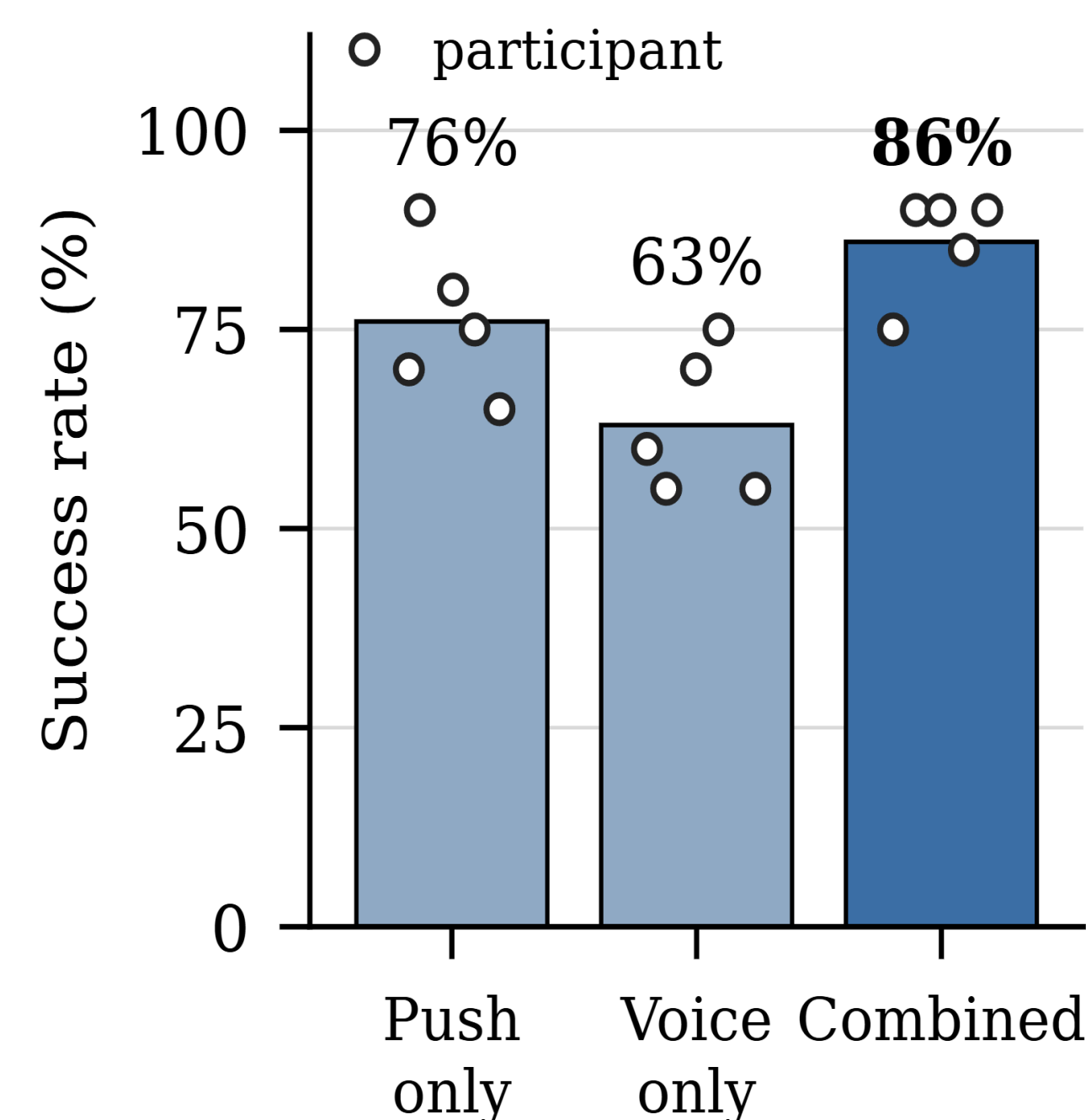
Algorithm Intent_Particle_Filter($\mathcal{X}_{k-1}, \mathbf{z}_k, \mathcal{S}$)

Input: particles $\mathcal{X}_{k-1} = \{\theta_{k-1}^{(n)}\}_{n=1}^N$
cue $\mathbf{z}_k = (\text{type}_k, \text{payload}_k)$
scene prior $\mathcal{S}$

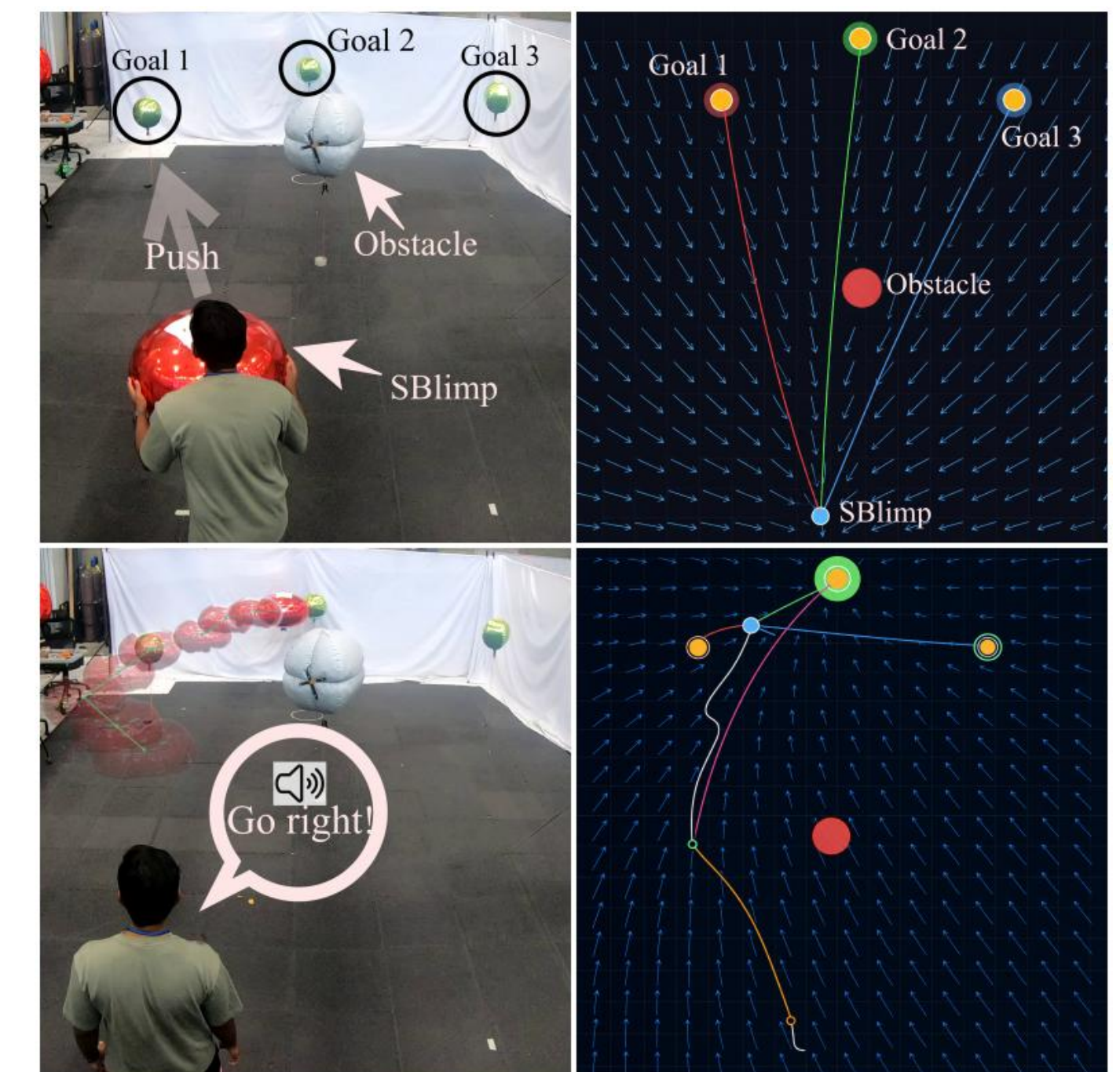
Output: $\mathcal{X}_k$, intent estimate θ_k^*

- 1: **for** $n = 1$ to N **do**
- 2: $\theta_k^{(n)} \leftarrow \theta_{k-1}^{(n)} + \eta^{(n)}$ $\triangleright$ predict
- 3: $w^{(n)} \leftarrow p(\mathbf{z}_k | \theta_k^{(n)})$ $\triangleright$ reweight
- 4: $w^{(n)} \leftarrow w^{(n)} / \sum_m w^{(m)} \forall n$ $\triangleright$ normalize
- 5: $i_k^* \leftarrow$ goal with the largest total weight
- 6: $\theta_k^* \leftarrow$ best particle of goal i_k^*
- 7: $\mathcal{X}_k \leftarrow$ resample $\propto w^{(n)}$ $\triangleright$ resample
- 8: **return** $\mathcal{X}_k, \theta_k^*$

Particle $\theta^{(n)} = (\vartheta_1^{(n)}, \vartheta_2^{(n)}, r^{(n)}, \mathbf{x}^{*(n)})$ is one LDS hypothesis with weight $w^{(n)}$; $\eta^{(n)}$ is zero-mean Gaussian noise on the shape components only. Goals are sampled from $\mathcal{S}$ at initialization and stay fixed per particle.



Experiment 1: The subject has to send the robot to one randomly chosen goal among 8 candidate goals. Combining modalities wins. 86% goal identification vs. 76% push-only and 63% voice-only. Circles are individual participants, combined is also the most consistent across users.



Experiment 2: Steering around an obstacle the robot can't see: The human wants Goal 2, but a straight line hits the obstacle. A push toward Goal 1 (top) leaves all goals plausible; "Go right!" (bottom) reweights toward Goal 2, and the executed trajectory curves around.